

Department of Mathematics and Computer Science

Study Guide for the course

Mathematics 2 (5EZB0)

for the Bachelor's program:

Electrical Engineering

Automotive Technology

Introduction

In the course Mathematics 2 (5EZB0), the notions of differentiation and integration that were introduced in the course Calculus (2WBB0) for functions of one variable are extended to the multi-variable case. Focus is on vector analysis in the 3-D case and the investigation of vector fields in this setting. The results have important applications in physics and engineering. For electrical engineers, a good understanding of the notion of vector fields and the ability to use appropriate techniques from vector analysis is indispensable in, for instance, the study of electro magnetics.

The course addresses the following topics:

- Curves, surfaces and parametrizations
- Functions of several variables; partial derivatives
- Multiple integration
- Vector fields, line and surface integrals, flux
- Notions of gradient, divergence and curl

This study guide describes the contents of the course Mathematics 2 (5EZB0). The course is based on parts of the following book:

R.A. Adams and C. Essex: Calculus – A complete course, Tenth Edition, Pearson Canada Inc., 2022. ISBN: 978-0-13-573258-8

This study guide is an addition to the book. First of all, it indicates which parts of the book will be discussed, and which exercises have to be made. If necessary, it will provide additional subject matter, consisting of both study material and additional exercises.

In the schedule on the next pages, you can find the dates of the **lectures** (in principle twice a week, but in weeks 2,3,4 and 6 only once, due to holidays), the **instructions** (once or twice a week) and the **studio classroom session** (in weeks 2,6,8). We expect that you **attend** all these educational activities on Monday mornings and Thursday afternoons.

In the remainder of this study-guide, the material is divided into 10 sections; each section describes the contents of one course week (we have 9 weeks during the quartile). Section 10 describes the details for the final test. Each section starts with a list of the educational activities in that week. After this list, (some of) the following subsections will appear:

1. Subjects

The first section of each chapter contains a list of the subjects to be treated from the **book** (ADAMS). You have to **read** the indicated sections in detail, and **study** them carefully. In the **lectures**, the subject material will be introduced and elaborated, to ensure a quick and smooth entry to the new material. The lectures will be recorded, so if necessary you can follow the lecture online.

2. Exercises

Here the exercises are described that we expect you to do in the corresponding week. They are divided into preparation exercises, that are rather basic, and that you should be able to do yourself at home, and instruction exercises, that are more challenging, and for which you probably need some guidance or help. To provide this assistance, there

Last name starts with	Room (see weekly description)	Instructor	TA
A until E	Group 1	Skantzaris	Job Perey
F until L	Group 2	Goering	Piotr Mikielewicz
M until R	Group 3	Hoeksema	Anna Tuzi
S until Z	Group 4	van Heumen	Yvan Bergmans

Table 1: Allocation of the students for the instruction classes

are 4 parallel instructions on Monday morning, from 10:45 until 12:30h, and –in case that there is no studio class room session– on Thursday from 15:30-17:15h.

During the instruction, you can ask questions on the theory and all exercises. We will try to answer your questions on an individual basis; for this reason there will be an additional student-assistant (TA) in each room. If there are many students with the same problem/question, the instructor will explain a solution on the blackboard. The instructors will also do selected problems on the blackboard every class.

In mastering mathematics, training your mathematical skills by making exercises is extremely important. For this reason, **your presence and attendance in the instructions is essential for a successful accomplishment of the course**. Our goal is to let you work independently on an individual basis or in a small group of students, and to provide help and additional explanations at the moment that you need it. If you attend the exercise classes on a regular basis, you will experience that this is a rather efficient way of studying and that it really helps you to master the subject.

Please note that the 2(+2) hours of instruction per week may not be enough to finish all weekly exercises. We expect that you need to spend 7 hours of self study per week, for studying the book, and to finish all exercises.

*In addition to the lectures and instructions, there will be **studio classroom sessions** on three Thursday afternoons. In these sessions you will work on problems related to automotive technology and/or electrical engineering, in which you will apply the mathematical techniques that are discussed in the course. This will mainly relate to the Physics for EE/AT course (5EPD0/5EPE0) that you took last quartile, or the Electromagnetics I course (5EPF0) that you will take next year. You will experience why mathematics is so important for automotive and electrical engineers, and at the same time improve both your engineering and mathematical skills. The problems that are discussed during this studio classroom session are not contained in this study guide. Instead they will be provided to you on a weekly basis through slide decks. There will be a question on a topic covered in a studio classroom session in the final exam.*

Assessment

The assessment of the course consists of three components.

- (i) **(Bi)-weekly electronic exercises.** The electronic exercises have to be made at home. We use the platform **SOWISO** for this. Starting in week 2, five small electronic tests will be made available in SOWISO, approximately on a bi-weekly basis. You need to log into the system, and make the weekly electronic tests before the indicated *deadline*.

On Canvas we will announce when a SOWISO test is made available, and the exact deadline for that test. In SOWISO, you can take the tests as many times as you want within the given timeframe. After each attempt, you will be able to see the answers for that attempt. After submitting your answers, your test will be automatically graded. The best grade for each test of all attempts will be the grade corresponding to that test. This grade will contribute to your grade for the electronic exercises: the weighted mean of your scores for all graded electronic tests determines 10% of your final grade for the course.

- (ii) **Midterm test** on Thursday 21 May 2026, 15:45-17:15 h. This will be test in ANS with mutiple choice and/or open exercises. The grade of the midterm determines 10% of your grade for the course. There are NO resits of the Midterm test.
- (iii) **Final test** on Monday 22 June 2026, 9:00-12:00 h. The final test is an on-campus written test in ANS. Your grade for the final test determines 80% of your grade for the course. An additional requirement for passing the course is that your grade for the final test must be at least 5.0. If your grade of the final test is less than 5.0 you fail the course, and your grade will be 'NVD'. In that case you can participate in the resit of the final test, which is scheduled on Monday 10 August 2026, 9:00-12:00 h.

The exact subject matter of the midterm and final test will be announced in due time.

Note: this study guide should be considered as work in progress and it is updated and extended during the quartile. Always check the latest version for your homework and tutorial exercises, information regarding additional study material, tests and exams, etc. If you notice typos or other mistakes in this study guide, please report them by e-mail to the lecturer of the course. All your help in improving this study guide is appreciated.

We hope that you will enjoy the course. On behalf of the Math 2 team,

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2025/2026

1 Week 1: Parametrization of curves and surfaces in 3-D

1.1 Educational activities

- Lecture 1.1: Monday 20 April 2026, 8:45–10:30 h. in Aud. 6
- Instruction 1: Monday 20 April 2026, 10:45–12:30 h. in Atlas -1.715 (Group 1), Atlas 4.215 (Group 2), Atlas 4.225 (Group 3), Atlas 6.225 (Group 4)
- Lecture 1.2: Thursday 23 April 2026, 13:30–15:15 h. in Aud. 6
- Instruction 2: Thursday 23 April 2026, 15:30–17:15 h. in Matrix 1.340 (Group 1), Matrix 1.345 (Group 2), Auditorium 16 (Group 3), Gemini-Noord 1.810 (Group 4)

1.2 Subjects

- ADAMS 10.1: Analytic geometry in three dimensions
- ADAMS 10.5: Quadric surfaces
- ADAMS 10.6: Cylindrical and spherical coordinates
- ADAMS 12.1: Vector functions of one variable
- ADAMS 12.3: Curves and parameterizations

1.3 Exercises

- ADAMS 10.1: Analytic geometry in three dimensions
 - Preparation: Exercises: 15,17,35
 - Instruction: Exercises: 18,19,22,27,28,36
- ADAMS 10.5: Quadric surfaces
 - Preparation: Exercises: 1,3,5
 - Instruction: Exercises: 7,8,15,17,19
- ADAMS 10.6: Cylindrical and spherical coordinates
 - Preparation: Exercises: 1,3
 - Instruction: Exercises: 6,9,13
- ADAMS 12.1: Vector functions of one variable
 - Preparation: Exercises: 3,9
 - Instruction: Exercises: 17,24
- ADAMS 12.3: Curves and parameterizations
 - Preparation: Exercises: 1,5
 - Instruction: Exercises: 3,9,13,17,19

2 Week 2: Functions of several variables; limits, continuity

2.1 Educational activities

- Lecture 2.1: Thursday 30 April 2026, 13:30–15:15 h. in Aud. 6
- Studio Classroom 1: Thursday 30 April 2026, 15:30–17:15 h. in Aud. 6

2.2 Subjects

- ADAMS 13.1: Functions of several variables
- ADAMS 13.2: Limits and continuity

2.3 Exercises

- ADAMS 13.1: Functions of several variables
 - Preparation: Exercises: 1,2,11,19
 - Instruction: Exercises: 6,7,14,15,17,23,24,33,39
- ADAMS 13.2: Limits and continuity
 - Preparation: Exercises: 1,3,4
 - Instruction: Exercises: 8,12,13

3 Week 3: Partial derivatives: first and higher order

3.1 Educational activities

- Lecture 3.1: Thursday 7 May 2026, 13:30–15:15 h. in Aud. 6
- Instruction 3: Thursday 7 May 2026, 15:30–17:15 h. in Matrix 1.340 (Group 1), Matrix 1.345 (Group 2), Auditorium 16 (Group 3), Gemini-Noord 1.810 (Group 4)

3.2 Subjects

- ADAMS 13.3: Partial derivatives
- ADAMS 13.4: Higher-order derivatives
- ADAMS 13.5: The chain rule

3.3 Exercises

- ADAMS 13.3: Partial derivatives
 - Preparation: Exercises: 1,4,15,17
 - Instruction: Exercises: 5,6,13,14,19,23,24,27 (in the underlined exercises, please give a vector description of the normal line, instead of two equations)
- ADAMS 13.4: Higher-order derivatives
 - Preparation: Exercises: 3,8,15
 - Instruction: Exercises: 1,4,5,11,12
- ADAMS 13.5: The chain rule
 - Preparation: Exercises: 1,3,13
 - Instruction: Exercises: 5,6,8,11,15,17

4 Week 4: Linear approximations, gradients and directional derivatives

4.1 Educational activities

- Lecture 4.1: Monday 11 May 2026, 8:45–10:30 h. in Aud. 6
- Instruction 4: Monday 11 May 2026, 10:45–12:30 h. in Atlas -1.715 (Group 1), Atlas 4.215 (Group 2), Atlas 4.225 (Group 3), Atlas 6.225 (Group 4)

4.2 Subjects

- ADAMS 13.6: Linear approximations, differentiability, and differentials
- ADAMS 13.7: Gradients and directional derivatives

4.3 Exercises

- ADAMS 13.6: Linear approximations, differentiability, and differentials
 - Preparation: Exercises: 1,3
 - Instruction: Exercises: 5,6,17,18,20
- ADAMS 13.7: Gradients and directional derivatives
 - Preparation: Exercises: 1,2,3,17
 - Instruction: Exercises: 4,5,7,11,14,19,22,26

5 Week 5: Taylor series; Introduction multiple integration – Double integrals

5.1 Educational activities

- Lecture 5.1: Monday 18 May 2026, 8:45–10:30 h. in Aud. 6
- Instruction 5: Monday 18 May 2026, 10:45–12:30 h. in Atlas -1.715 (Group 1), Atlas 4.215 (Group 2), Atlas 4.225 (Group 3), Atlas 6.225 (Group 4)
- Lecture 5.2: Thursday 21 May 2026, 13:30–15:15 h. in Aud. 6
- **INTERMEDIATE TEST:** Thursday 21 May 2026, 15:45–17:15 h. in rooms to be announced. The subject material will be what is stated in the study guide up to and including week 4.

5.2 Subjects

- ADAMS 13.9: Taylor's formula, Taylor series, and approximations
- ADAMS 15.1: Double integrals
- ADAMS 15.2: Iteration of double integrals in Cartesian coordinates

5.3 Exercises

- ADAMS 13.9: Taylor's formula, Taylor series, and approximations
 - Preparation: Exercises: 1,2,3 (in all exercises only Taylor polynomial of degree 2)
 - Instruction: Exercises: 4,5,6,8,9 (in exercises 5 and 6 only Taylor polynomial of degree 2)
- ADAMS 15.1: Double integrals
 - Preparation: Exercises: 13,14
 - Instruction: Exercises: 16,17,19,20
- ADAMS 15.2: Iteration of double integrals in Cartesian coordinates
 - Preparation: Exercises: 1,2,5
 - Instruction: Exercises: 7,9,10,13,15,17,21,27

6 Week 6: Double integrals in polar coordinates; Triple integrals

6.1 Educational activities

- Lecture 6.1: Thursday 28 May 2026, 13:30–15:15 h. in Aud. 6
- Studio Classroom 2: Thursday 28 May 2026, 15:30–17:15 h. in Aud. 6

6.2 Subjects

- ADAMS 15.4: Double integrals in polar coordinates
- ADAMS 15.5: Triple integrals

6.3 Exercises

- ADAMS 15.4: Double integrals in polar coordinates
 - Preparation: Exercises: 1,3,8,19
 - Instruction: Exercises: 6,9,12,13,16,21,23
- ADAMS 15.5: Triple integrals
 - Preparation: Exercises: 1,3,5,17 or 21.
 - Instruction: Exercises: 7,9,14,19 or 23,27

7 Week 7: Change of variables in triple integrals; Introduction to vector fields

7.1 Educational activities

- Lecture 7.1: Monday 1 June 2026, 8:45–10:30 h. in Aud. 6
- Instruction 6: Monday 1 June 2026, 10:45–12:30 h. in Atlas -1.715 (Group 1), Atlas 4.215 (Group 2), Atlas 4.225 (Group 3), Atlas 6.225 (Group 4)
- Lecture 7.2: Thursday 4 June 2026, 13:30–15:15 h. in Aud. 6
- Instruction 7: Thursday 4 June 2026, 15:30–17:15 h. in Atlas -1.310 (Group 1), Matrix 1.345 (Group 2), Auditorium 16 (Group 3), Gemini-Noord 1.810 (Group 4)

7.2 Subjects

- ADAMS 15.6: Change of variables in triple integrals
- ADAMS 15.7: Applications of multiple integrals (only "The surface area of a graph")
- ADAMS 16.1: Vector and scalar fields (not including "Lyapunov functions")
- ADAMS 16.2: Conservative fields

7.3 Exercises

- ADAMS 15.6: Change of variables in triple integrals
 - Preparation: Exercises: 2, 3, 5
 - Instruction: Exercises: 6, 8, 11, 13, 16, 18
- ADAMS 15.7: Applications of multiple integrals (only "The surface area of a graph")
 - Preparation: Exercises: 1, 5, 8
 - Instruction: Exercises: 10,11,12
- ADAMS 16.1: Vector and scalar fields
 - Preparation: Exercises: 1, 2, 3, 4, 7
 - Instruction: Exercises: 10, 12, 13, 14, 16, 20
- ADAMS 16.2: Conservative fields
 - Preparation: Exercises: 1, 3, 4, 5
 - Instruction: Exercises: 6, 9, 11, 20

8 Week 8: Line and surface integrals

8.1 Educational activities

- Lecture 8.1: Monday 8 June 2026, 8:45–10:30 h. in Aud. 6
- Instruction 8: Monday 8 June 2026, 10:45–12:30 h. in Atlas -1.715 (Group 1), Atlas 4.215 (Group 2), Atlas 4.225 (Group 3), Atlas 6.225 (Group 4)
- Lecture 8.2: Thursday 11 June 2026, 13:30–15:15 h. in Aud. 6
- Studio Classroom 3: Thursday 11 June 2026, 15:30–17:15 h. in Aud. 6

8.2 Subjects

- ADAMS 16.3: Line integrals
- ADAMS 16.4: Line integrals of vector fields
- ADAMS 16.5: Surfaces and surface integrals
- ADAMS 16.6: Oriented surfaces and flux integrals
- ADAMS 17.1: Gradient, divergence and curl (NOT Distributions and delta functions)

8.3 Exercises

- ADAMS 16.3: Line integrals
 - Preparation: Exercises: 1, 2, 3, 4, 5
 - Instruction: Exercises: 6, 7, 10, 11, 17
- ADAMS 16.4: Line integrals of vector fields
 - Preparation: Exercises: 2, 4, 5, 6, 8
 - Instruction: Exercises: 10, 14, 16, 19, 21.
 - Additional interesting problems: Exercises 20, 22, 23, 24
- ADAMS 16.5: Surfaces and surface integrals
 - Preparation: Exercises: 2, 3, 4, 5
 - Instruction: Exercises: 6, 9, 14, 17, 22, **hard**: 18, 20
- ADAMS 16.6: Oriented surfaces and flux integrals
 - Preparation: Exercises: 1, 2, 3, 4
 - Instruction: Exercises: 5, 7, 11, 12, 15, 17
- ADAMS 17.1: Gradient, divergence and curl (NOT Distributions and delta functions)
 - Preparation: Exercises: 1, 2, 4, 6
 - Instruction: Exercises: 7, 8, 10, 11, 12

9 Week 9: Repetition and exam preparation

9.1 Educational activities

- Lecture 9.1: Monday 15 June 2026, 8:45–10:30 h. in Aud. 6
- Instruction 9: Monday 15 June 2026, 10:45–12:30 h. in Atlas -1.715 (Group 1), Atlas 4.215 (Group 2), Atlas 4.225 (Group 3), Atlas 6.225 (Group 4)
- Lecture 9.2: Thursday 18 June 2025, 13:30–15:15 h. in Aud. 6
- Instruction 10: Thursday 18 June 2026, 15:30–17:15 h. in Matrix 1.340 (Group 1), Matrix 1.345 (Group 2), Auditorium 16 (Group 3), Gemini-Noord 1.810 (Group 4)

9.2 Subjects

In week 9, no new concepts will be introduced, unless some topics from the previous weeks were delayed (possible due to many holidays in between). In that case, these topics from previous weeks will be covered in week 9. During the lectures and instructions the more difficult subjects from the previous weeks will be repeated, and also a significant amount of time will be devoted for preparation for the final test.

10 Final test

The final test is scheduled on

- Monday 22 June 2026, 9:00–12:00 h.

There is a resit for this final test in the interim period:

- Monday 10 August 2026, 9:00–12:00 h.

The subject material consists of all the sections from the book of ADAMS that were covered during the course.

1. Chapter 10: Sections 10.1, 10.5 and 10.6
2. Chapter 12: Sections 12.1 and 12.3
3. Chapter 13: Sections 13.1 until 13.7 and 13.9
4. Chapter 15: Sections 15.1, 15.2, and 15.4, 15.5, 15.6, 15.7(only 'The surface area of a graph')
5. Chapter 16: Sections 16.1 until 16.6
6. Chapter 17: Sections 17.1 (without 'distributions and delta functions')

If there are changes in the subject material, then they will be announced in due time on the Canvas page of the course.